



Reading a Genetic Analysis Report

Genetics GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 4 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.

EASY

In the notation 'c.1234C>T', what does the 'c.' prefix indicate?

- A. A chromosomal location
- B. A protein-level change
- C. A change at the coding DNA sequence level
- D. The exon number
- E. The zygosity of the variant

2.

EASY

How many variant classification tiers were taught in this lecture?

- A. Three (Pathogenic, Likely pathogenic, VUS)
- B. One
- C. Two
- D. Five
- E. Four

3.

EASY

A Variant of Uncertain Significance (VUS) means:

- A. The variant is definitely benign
- B. There is insufficient evidence to classify the variant as benign or pathogenic
- C. The variant is confirmed to cause disease
- D. The variant is always inherited
- E. The variant was not sequenced

4.

EASY

Which section of a genetic test report describes the DNA extraction method, sequencing method, and targeted genes?

- A. Recommendations
- B. Clinical interpretation
- C. Patient information
- D. Limitations
- E. Methodology

5.

MEDIUM

A variant has strong evidence supporting pathogenicity, but not yet enough data for a definitive classification. How should it be classified?

- A. Likely benign
- B. Pathogenic
- C. Likely pathogenic
- D. VUS
- E. Benign

6.

MEDIUM CASE

A patient is found to carry one pathogenic PKHD1 variant (heterozygous), consistent with carrier status rather than disease.

What is the most appropriate next recommendation for this patient?

- A. Genetic counselling regarding carrier status and family planning
- B. Immediate liver transplant evaluation
- C. No further action needed at any point
- D. Empirical antiviral therapy
- E. Repeat the identical test in 6 months

7.

MEDIUM

A genetic panel returns negative for all analysed genes in a patient with unexplained liver disease. What is the correct interpretation?

- A. The patient definitely does not have a genetic liver disease
- B. The test was invalid and must be repeated immediately
- C. The patient is a confirmed carrier
- D. The likelihood of the tested condition is reduced, but other causes are not ruled out
- E. The result proves an acquired, non-genetic cause

8.

MEDIUM

In the worked example, a homozygous pathogenic PKHD1 variant with nucleotide change c.1234C>T and protein change p.(Arg412Trp) is reported. What does 'homozygous' indicate about clinical significance here?

- A. The variant is absent from the genome
- B. The patient is an unaffected carrier only
- C. The variant is present on both alleles and supports the clinical diagnosis of disease in this patient
- D. Homozygosity always indicates a benign polymorphism
- E. The finding is of uncertain significance regardless of zygosity

9.

MEDIUM

Which section of a genetic report specifically states what the test CANNOT rule out?

- A. Referring physician
- B. Clinical interpretation
- C. Test details
- D. Results
- E. Limitations

10.

MEDIUM

In 'c.1234C>T', what does the 'C>T' portion specifically represent?

- A. A substitution of cytosine (original) with thymine (new) at that position
- B. A chromosomal translocation
- C. An insertion between positions 1234 and 1235
- D. A deletion of the cytosine base
- E. A duplication of the thymine base

11.

MEDIUM

A 'positive' result on a genetic test report means:

- A. The patient is a carrier only, with no disease risk
- B. The test could not be interpreted
- C. The condition is definitely ruled out
- D. No pathogenic variants were found
- E. A mutation associated with the disease was identified, supporting the clinical diagnosis and a genetic basis for the condition

12.

MEDIUM

Why does this lecture describe the PKHD1/congenital hepatic fibrosis example as a 'template case'?

- A. Because it is the only possible genetic cause of liver fibrosis
- B. Because PKHD1 mutations are the most common cause of all liver disease
- C. Because the report format changes depending on the disease
- D. Because PKHD1 is unrelated to the report-reading skill being taught
- E. Because the lecture is teaching the structure of a genetic report using this case as a worked example, not teaching congenital hepatic fibrosis as a disease topic

13.

MEDIUM

Which of the following would appropriately appear in the 'Recommendations' section of a genetic test report?

- A. The bioinformatics tools used for analysis
- B. Referral to a specialist, genetic counselling, and/or further testing
- C. The exact nucleotide change identified
- D. The name of the referring physician only
- E. The patient's date of birth

14.

MEDIUM

A report lists a protein change as 'p.(Arg412Trp)'. This notation describes a change at which level?

- A. The mRNA splicing level only
- B. The coding DNA sequence level
- C. The protein sequence level, showing an amino acid substitution
- D. The epigenetic (methylation) level
- E. The chromosomal level

15.

HARD

CASE

A clinician receives a report classifying a variant as 'Likely pathogenic' and wants to know how confidently to act on it compared to a 'Pathogenic' classification.

What is the key practical difference between these two classifications?

- A. 'Likely pathogenic' reflects strong but not yet definitive evidence, whereas 'Pathogenic' is confirmed to cause the disease with sufficient evidence
- B. 'Pathogenic' requires no further clinical correlation while 'Likely pathogenic' does not exist in modern reporting
- C. 'Likely pathogenic' only applies to somatic, not germline, variants
- D. They are functionally identical and require the same action
- E. 'Likely pathogenic' means the variant is benign

16.

HARD

CASE

A family requests interpretation of a relative's genetic report, which shows a heterozygous pathogenic variant in a gene associated with an autosomal recessive liver condition, with no symptoms in the tested relative.

Based on the classification framework in this lecture, how should this heterozygous finding be explained to the family?

- A. The relative has the disease and needs immediate treatment
- B. The relative is a carrier — unaffected themselves, but the finding is relevant for family planning and relatives' testing
- C. The variant must be reclassified as benign since the relative is asymptomatic
- D. Carrier status always progresses to clinical disease later in life
- E. The result is invalid because the relative shows no symptoms

17.

HARD

A report shows a negative result for a targeted gene panel in a patient with a strong clinical picture of a genetic liver disease. Which limitation of genetic testing best explains why the diagnosis cannot be fully excluded?

- A. A negative result only rules out variants within the specific genes/regions analysed, not all possible genetic or non-genetic causes
- B. A negative result confirms an acquired, non-genetic diagnosis
- C. Negative results are always false and should be disregarded
- D. The report format does not allow for negative results
- E. Negative results always indicate poor sample quality

18.

HARD

Why does a clinical genetic report separate 'Results' (gene, nucleotide/protein change, zygosity, classification) from 'Clinical interpretation' (what a positive/carrier/negative result means in plain language)?

- A. Because results are optional but interpretation is mandatory
- B. Because the two sections always contradict each other
- C. Because the technical variant data and its plain-language clinical meaning serve different audiences and purposes, and separating them prevents misreading a technical result as a diagnosis or vice versa
- D. Because only the interpretation section is medically valid
- E. Because zygosity is never relevant to interpretation

19.

HARD

CASE

A genetic counsellor is asked why the sequencing methodology section (NGS vs Sanger, bioinformatics tools) matters when interpreting a VUS finding.

What is the best justification?

- A. NGS results override any classification made by the report
- B. Sanger sequencing can never detect a VUS
- C. Methodology is irrelevant once a classification is assigned
- D. VUS findings are always methodology-independent artifacts
- E. The sequencing method and analysis pipeline can affect variant detection sensitivity and confidence, which is relevant context when a variant's significance is still uncertain

20.

HARD

Given the three classification tiers (Pathogenic, Likely pathogenic, VUS) presented in this lecture, which statement best reflects appropriate clinical caution?

- A. All three tiers should be acted upon identically
- B. A VUS should not be used to make a definitive diagnosis or guide major clinical decisions the way a Pathogenic or Likely pathogenic result might
- C. A VUS is equivalent to a Pathogenic result for reproductive counselling purposes
- D. Only Likely pathogenic variants require any follow-up
- E. Pathogenic variants require less action than VUS findings

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	C	A change at the coding DNA sequence level
2	A	Three (Pathogenic, Likely pathogenic, VUS)
3	B	There is insufficient evidence to classify the variant as benign or pathogenic
4	E	Methodology
5	C	Likely pathogenic
6	A	Genetic counselling regarding carrier status and family planning
7	D	The likelihood of the tested condition is reduced, but other causes are not ruled out
8	C	The variant is present on both alleles and supports the clinical diagnosis of disease in this patient
9	E	Limitations
10	A	A substitution of cytosine (original) with thymine (new) at that position
11	E	A mutation associated with the disease was identified, supporting the clinical diagnosis and a genetic basis for the condition
12	E	Because the lecture is teaching the structure of a genetic report using this case as a worked example, not teaching congenital hepatic fibrosis as a disease topic
13	B	Referral to a specialist, genetic counselling, and/or further testing
14	C	The protein sequence level, showing an amino acid substitution
15	A	'Likely pathogenic' reflects strong but not yet definitive evidence, whereas 'Pathogenic' is confirmed to cause the disease with sufficient evidence
16	B	The relative is a carrier — unaffected themselves, but the finding is relevant for family planning and relatives' testing
17	A	A negative result only rules out variants within the specific genes/regions analysed, not all possible genetic or non-genetic causes
18	C	Because the technical variant data and its plain-language clinical meaning serve different audiences and purposes, and separating them prevents misreading a technical result as a diagnosis or vice versa
19	E	The sequencing method and analysis pipeline can affect variant detection sensitivity and confidence, which is relevant context when a variant's significance is still uncertain
20	B	A VUS should not be used to make a definitive diagnosis or guide major clinical decisions the way a Pathogenic or Likely pathogenic result might

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Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.